Determinants of Household Participation in Securities Markets: An Empirical Analysis

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***Abstract*—Household participation in capital markets is a crucial driver of financial inclusion and wealth creation, yet it remains disproportionately skewed across demographic lines. Using a large-scale micro-dataset from the 2025 SEBI Household Survey ($N = 109,430$), this paper empirically analyzes the determinants of securities market participation in India. We employ a logistic regression framework to isolate the socioeconomic drivers of investment, revealing that education and income are the strongest predictors (odds ratios of 1.83 and 1.80, respectively). We further investigate structural barriers deterring non-investors, identifying loss aversion and information asymmetry as primary friction points. To deepen our behavioral insights, we expose a stark Dunning-Kruger effect among overconfident retail traders and measure the impact of "finfluencers" on high-risk asset adoption. Finally, we implement Random Forest classifiers to predict participation, asset choice, and investment horizons based on basic demographic metrics. Our predictive models achieve strong discriminatory power for baseline participation (AUC=0.76) but reveal that predicting the specific nature of retail investments requires deeper psychographic data, providing actionable insights for targeted regulatory interventions.**

***Index Terms*—Financial Inclusion, Securities Markets, Household Finance, Behavioral Economics, Predictive Modeling**

I. Introduction

The democratization of capital markets is essential for broad-based economic growth and household wealth accumulation. In emerging economies like India, retail participation in securities has witnessed a significant surge, catalyzed by digitization and the proliferation of zero-commission trading platforms. Despite this growth, participation remains highly heterogeneous.

Understanding the determinants of household participation is a central problem in financial economics. Why do certain households allocate capital to mutual funds and equities, while others restrict their savings to traditional, low-yield instruments like fixed deposits? The literature points to a combination of background risk, financial literacy, and trust in institutions. However, most existing studies rely on older data that predate the post-pandemic retail boom, the widespread adoption of digital KYC, and the systemic rise of social media financial influencers.

This paper addresses this gap by utilizing the comprehensive 2025 SEBI Household Survey. We pose four core research questions: (1) Who is likely to participate in the securities market? (2) What structural and behavioral barriers keep the majority of households out? (3) Which securities do participating households prefer, and for how long? and (4) Can we predict these outcomes given basic demographic and behavioral metrics?

The remainder of the paper is organized as follows: Section II details the dataset and methodology. Section III presents the empirical findings regarding determinants, behavioral anomalies, and securities preferences. Section IV details our predictive modeling framework. Finally, Section V concludes with highly condensed policy implications.

II. Data and Methodology

A. Dataset

We utilize micro-data from the 2025 SEBI Household Survey on Securities Market Participation, comprising a nationally representative sample of 109,430 households across India. The dataset captures detailed demographic variables (age, gender, education, income), geographical data, and behavioral metrics (risk tolerance, market perception). Crucially, the survey differentiates between *investors* (25.5%) and *non-investors* (74.5%).

B. Econometric & Predictive Specification

To identify the core determinants of participation, we first estimate a binary logistic regression model where the dependent variable is active investment status ($Y_i \in \{0, 1\}$). Independent variables are normalized to extract comparable odds ratios.

To expand this into a forecasting paradigm, we utilize Random Forest classifiers to predict three distinct outcomes: (Model 1) the binary likelihood of participation; (Model 2) the choice between market-linked securities (Equity/MF) vs. traditional instruments; and (Model 3) the propensity to hold equity for the long-term.

III. Empirical Results

A. Determinants of Participation & Overconfidence

The results of the logistic regression demonstrate a stark divide driven by human and financial capital. As depicted in Fig. 1, education and income emerge as the strongest statistical determinants. A one-standard-deviation increase in years of

Based on the 2025 SEBI Household Survey on Securities Market Participation.

education increases the odds of market participation by 1.83 times, holding other factors constant. Similarly, the odds ratio for log-income is 1.80. Urban residency (OR=1.56) and male gender (OR=1.56) also play significant roles, highlighting systemic geographical and gender-based disparities.

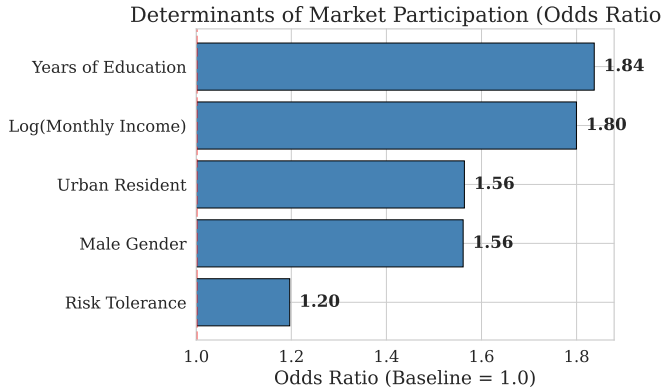

Fig. 1. Forest plot of Odds Ratios for determinants of market participation. Baseline is 1.0. Education and income show the highest predictive power.

For the 74.5% of households that remain completely outside the securities market, self-reported barriers (Fig. 2) indicate that behavioral frictions dominate structural ones. Specifically, "Fear of Loss" is the top barrier, closely followed by "Lack of Knowledge" and "Lack of Trust". This suggests that while digital platforms have solved access friction, the cognitive and emotional friction of investing remains high.

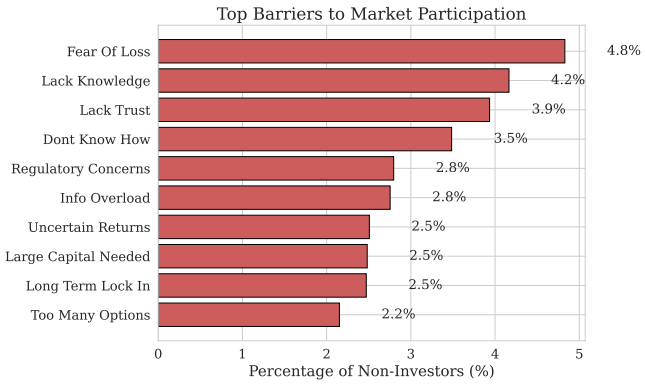

Fig. 2. Top barriers cited by non-investors. Loss aversion and information asymmetry (lack of knowledge/trust) are the primary deterrents.

Diving deeper into the "Knowledge" barrier, we analyzed the divergence between self-reported market familiarity and actual scores on an objective financial literacy quiz. We discovered a profound Dunning-Kruger effect: investors who are "overconfident" (high perceived familiarity, low actual knowledge) are disproportionately likely to hold high-risk assets like Futures & Options (F&O) and Cryptocurrency compared to calibrated investors (Fig. 3). This behavioral anomaly underscores why traditional models of rational participation often fail to capture modern retail dynamics.

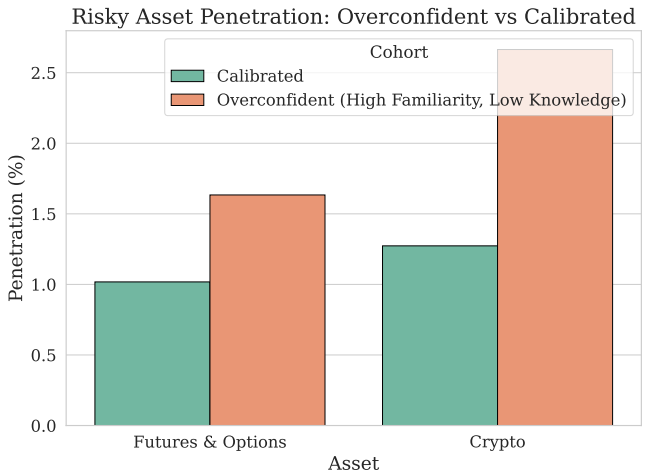

Fig. 3. Risky Asset Penetration. Overconfident investors participate in highly speculative assets at significantly higher rates than calibrated investors.

B. Securities Preferences, Demographics, and Influencers

Once households overcome the barrier to entry, their product preferences shift dynamically with income. Fig. 4 illustrates the penetration of various financial instruments across income quartiles. Fixed Deposits (FDs) remain the bedrock of Indian savings, showing uniformly high penetration. However, the adoption of market-linked securities—specifically Mutual Funds/ETFs and Direct Equity—scales dramatically in the higher income brackets, indicating that risk appetite scales non-linearly with wealth.

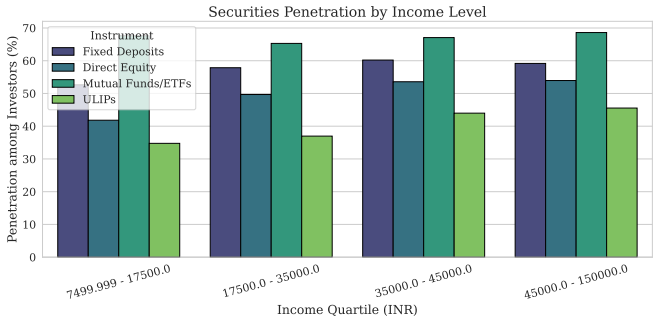

Fig. 4. Securities penetration among investors, segmented by income quartile. Market-linked instruments scale sharply with wealth.

Crucially, the choice of securities is heavily modulated by where investors source their information. Contrasting investors who rely exclusively on Social Media "Finfluencers" against those using traditional Financial Professionals reveals a sharp behavioral divide. As seen in Fig. 5, the Finfluencer cohort is primarily motivated by "Quick Gains," whereas the professionally-advised cohort seeks "Long Term Growth." This ties directly to the overconfidence trap discussed earlier, painting a picture of a segmented market where information source dictates risk behavior.

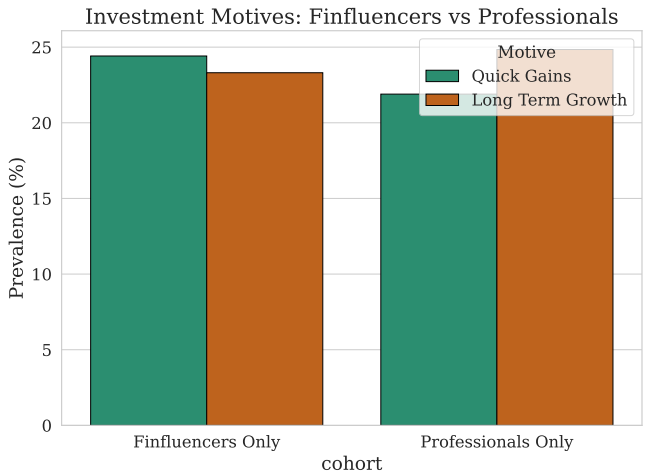

Fig. 5. Investment Motives segmented by Information Source. Social media heavily biases investors toward short-term speculative gains.

C. Investment Horizons

Holding duration preferences (Fig. 6) challenge the notion that all retail investors are short-term speculators. A significant plurality of equity and mutual fund investors define their holding periods as "Long term". In contrast, sophisticated derivative products (F&O) are predominantly utilized for "Short term" horizons, aligning with the motivations of the Finfluencer-driven cohort.

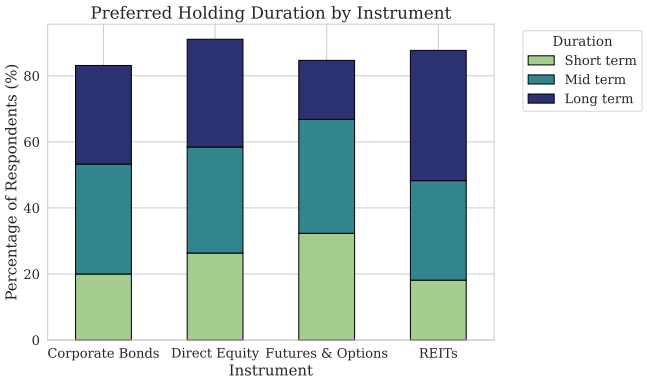

Fig. 6. Preferred holding durations across different asset classes. Direct equities are largely held for the long term, whereas derivatives skew short term.

IV. Predictive Modeling: Forecasting Household Participation

To operationalize our findings, we tasked Random Forest classifiers to predict household behavior based strictly on basic demographic and risk-tolerance metrics. Can we forecast what a household will do before they even open a brokerage account?

As shown in Fig. 7, the models yielded illuminating results regarding the limits of demographic forecasting:

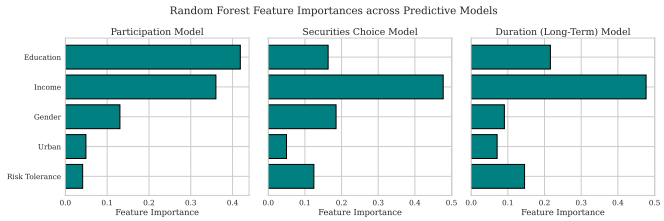

Fig. 7. Feature Importance across the three Predictive Models. Education and Income dominate the primary participation decision.

1) **Model 1: Predicting Participation (AUC = 0.76)**: Basic metrics are highly effective at predicting *whether* a household will participate in the market. Education and Income are the dominant features, confirming our logistic regression findings.
2) **Model 2: Predicting Securities Choice (AUC = 0.60)**: Predicting *how* they participate (choosing Equities vs. sticking to traditional FDs) is significantly harder. Once the wealth barrier is crossed, demographics lose their predictive power, and unobserved variables (like Finfluencer exposure) likely take over.
3) **Model 3: Predicting Investment Horizon (AUC = 0.58)**: Forecasting *how long* they will hold equity based on demographics is nearly indistinguishable from random chance.

These predictive limits validate our earlier behavioral findings: while structural socioeconomic factors gatekeep entry into the market, post-entry behavior is overwhelmingly dictated by psychological factors, overconfidence, and modern social media dynamics.

V. Conclusion and Policy Implications

The 2025 SEBI Household Survey provides compelling evidence that while the Indian securities market is growing, entry remains fundamentally gated by education and income. Once inside the market, however, traditional demographics fail to predict behavior. Instead, a dangerous cocktail of overconfidence and social media influence is driving a subset of retail investors toward highly speculative, short-term derivative trading.

For policymakers, the mandate is clear: interventions must pivot from merely reducing transaction costs to aggressively policing the quality of financial information. Bridging the knowledge gap via targeted, gamified financial literacy programs is essential. Furthermore, the undeniable Dunning-Kruger effect suggests that access to complex derivatives should perhaps be gated by objective literacy tests, rather than self-reported familiarity, to protect the new wave of retail investors from behavioral pitfalls.

References

[1] SEBI, "Household Survey on Securities Market Participation 2025," Securities and Exchange Board of India, 2025.